

Antis who are concerned about energy consumption in AI art. Why don't you care about 4k video streaming energy consumption? 80% of electricity consumed by the internet is caused by video streaming

Posted by u/SpeedFarmer42 • • 0 points • 66 comments

I posted this as a comment originally, but I thought it was worth discussing on its own.

4K video streaming uses enormous amounts of electricity, far more than AI image generation. I don't hear anyone complaining about that. Arguably 1080p is more than good enough IMO.

The European average is 56 grams of CO2 emissions per hour of video streaming. For comparison: 100 meters to drive causes 22 grams of CO2.

[<https://www.ndc-garbe.com/data-center-how-much-energy-does-a-stream-consume/>]
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80 percent of the electricity consumption on the Internet is caused by streaming services

Telekom needs the equivalent of 91 watts for a gigabyte of data transmission.

An hour of video streaming needs more than three times more energy than a HD stream in 4K quality, according to the Borderstep Institute. On a 65-inch TV, it causes 610 grams of CO2 per hour.

[<https://www.handelsblatt.com/unternehmen/it-medien/netflix-disney-und-co-klimakiller-streaming-so-koennen-sie-energie-beim-filmeschauen-einsparen/29410674.html>]
(<https://www.handelsblatt.com/unternehmen/it-medien/netflix-disney-und-co-klimakiller-streaming-so-koennen-sie-energie-beim-filmeschauen-einsparen/29410674.html>)

Comments

u/SingleProtection2501 • 1 points • 2024-12-31 23:09 UTC

I personally do care about the environment, enough to stream 480p on my computer devices and disconnect my phone from internet when I'm not using. I sadly can't say that about a lot of other antis. I'm more concerned about the water consumption though the carbon impact is also something to mind.

Now to get more subjective. I think that art should be an expression of emotions, a way of taking your inner feelings and getting them out, I personally make music as a way of coping with my depression as an example. I'd use AI for visualising a D&D character or something of the like, as I wouldn't necessarily call that an expression of your deepest emotions and thoughts, but in the large majority of cases it isn't used for depicting D&D characters.

u/Relative-Pen4046 • 2 points • 2024-12-31 12:03 UTC

As an electrical engineer going into the power industry, it's not the concern of Antis or Pro AI people to worry about the demands of our grid.

u/[deleted] • 1 points • 2024-12-31 11:35 UTC

[removed]

u/Elven77AI • 1 points • 2024-12-31 11:29 UTC

It would take weeks of photography, lots of fuel(both for planes and cars) and food(for photographer), to replicate a single AI batch of "nature landscape" photographs done in a half-hour.

That is for one location, you will need to buy plane tickets to another or just change the prompt? AI saves Terrawatts equivalent of energy and is much cheaper for the consumer(much less time, equipment and effort)

u/soerenL • -7 points • 2024-12-31 10:21 UTC

Who are you guys arguing with ?

u/Aphos • 3 points • 2024-12-31 13:56 UTC

Sure, it's possible, but if you're concerned about X because it's going to kill the planet, it is only intellectually honest to be concerned about Y because it's also going to kill the planet. So, if you're worried about AI killing the planet, than other digital activities that are also doing it and worse should also be on the chopping block.

u/soerenL • -1 points • 2024-12-31 14:38 UTC

What is it specifically that makes you think that ppl that are concerned with the energy consumption of gen AI datacenters, are not concerned with other things that are also damaging to our ecosystem ?

u/Weary-Swimming1432 • 1 points • 2025-01-01 15:48 UTC

Because they're using them.

u/soerenL • 1 points • 2025-01-01 16:32 UTC

What are they using, who are they, and how do you know ?

u/lovestruck90210 • 1 points • 2024-12-31 09:26 UTC

I can't find anything to corroborate the idea that "80 percent of electricity consumption on the internet is caused by streaming services". That'd be pretty major if true. From the Handelsblatt article you linked:

> 80 percent of electricity consumption on the Internet is caused by streaming services, says Ralph Hintemann, who researches the sustainability of digitalization at the Borderstep Institute, a Berlin think tank.

It'd be nice if Mr. Hintemann quoted the source for this number. I can't really seem to find any studies saying anything like that, not even those written by him. What I *do* see is that in the blog post you linked from NDC-Garbe, it says:

> According to experts, the operation of streaming services accounts for a significant proportion of **global data traffic**, around 80 percent in Germany.

It's talking about **data traffic**, not **electricity usage**. Totally different things. Funnily enough, it links back to the same Handelsblatt article as the source.

After doing some digging, this figure seems to originate from [a study done by the Shift Project] (<https://theshiftproject.org/wp-content/uploads/2019/07/2019-02.pdf>) in 2019 which said:

> Among these contents, videos make up 80% of world traffic and 80% of their growth in volume.

This pretty much cements my assumption that the 80% figure is referring to data traffic and NOT electricity consumption.

While we are on the topic of electricity consumption though, a [recent Goldman Sachs report] (<https://www.goldmansachs.com/pdfs/insights/pages/generational-growth-ai-data-centers-and-the-coming-us-power-surge/report.pdf>) predicts that AI will account for 19% of the power consumption by data centers by 2028.

> Our base case implies data center power demand moves from 1%-2% of overall global power demand to 3%-4% by 2030. The increase in the US is even greater — from 3% to 8%. Our estimates for overall data center power demand are above IEA forecasts (2026), and our outlook for AI to represent about 19% of data center power demand in 2028 is above recent corporate forecast.

A [report by Schneider Electric](https://www.se.com/ww/en/download/document/SPD_WP110_EN/) predicts that AI will account for 15-20% power consumption by 2028.

So yeah, AI is predicted to account for close to close to a fifth of all data center power consumption over the next 3 years.

u/zixaphir • 2 points • 2025-01-01 05:23 UTC

Unfortunately it seems such a basic criticism as "did you read your own source?" is enough to get you downvoted here.

u/soerenL • 2 points • 2024-12-31 13:35 UTC

Thanks for taking the time to look it up. To add to this: I haven't heard about any streaming services with plans to build nuclear power plants, in order to power streaming.

<https://www.bbc.com/news/articles/c748gn94k95o.amp>

u/Aphos • 2 points • 2024-12-31 13:52 UTC

well, yes, there are plans to build non-polluting energy sources for AI, thus lowering AI's carbon footprint. It would kind of rule if Netflix or Disney put some money towards this sort of thing too, actually.